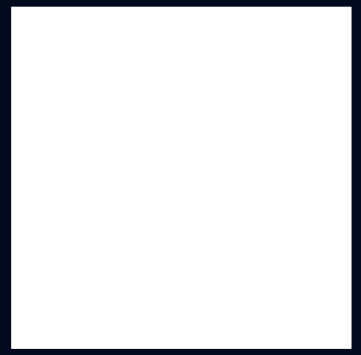


Ambient Systems

Climate Management for Cities of Tomorrow

Ambient Systems

An international team consisting of experienced entrepreneurs, engineers, and management consultants with a wealth of experience across major markets.



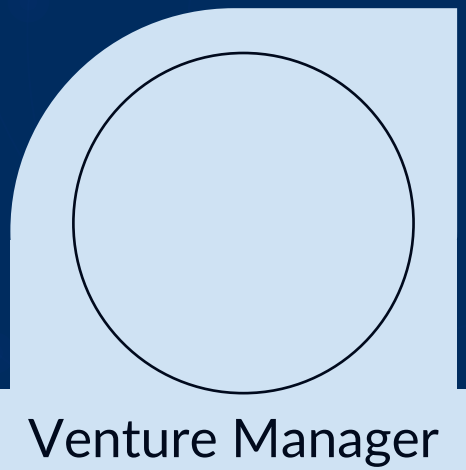
Product Development

Computer Science Master's Student
NUS (Singapore)



Corporate Development

Raised over $23M for ventures in highly regulated industries in Canada & EU



Venture Manager

Advisory

Established track-record of mentoring successful deep-tech spin-offs



Commercial Champion

Advisory

Experienced business operator of large enterprises and startups in APAC and China

Modern Buildings
Face Many Challenges

1. Climate Change
2. Urban Pollution
3. Indoor Contaminants
4. Rising Energy Costs
5. Increasing Technological Complexity



Indoor Farming Opportunity

1. Vertical Farming is a rapidly growing industry with 24% CAGR*.

2. Asia-Pacific is the global leader in vertical farming with market value of >$350M* in 2015.

3. Clear need from several farm operators in Singapore.

*Source: Statista

The Urban Farmer's Dilemma





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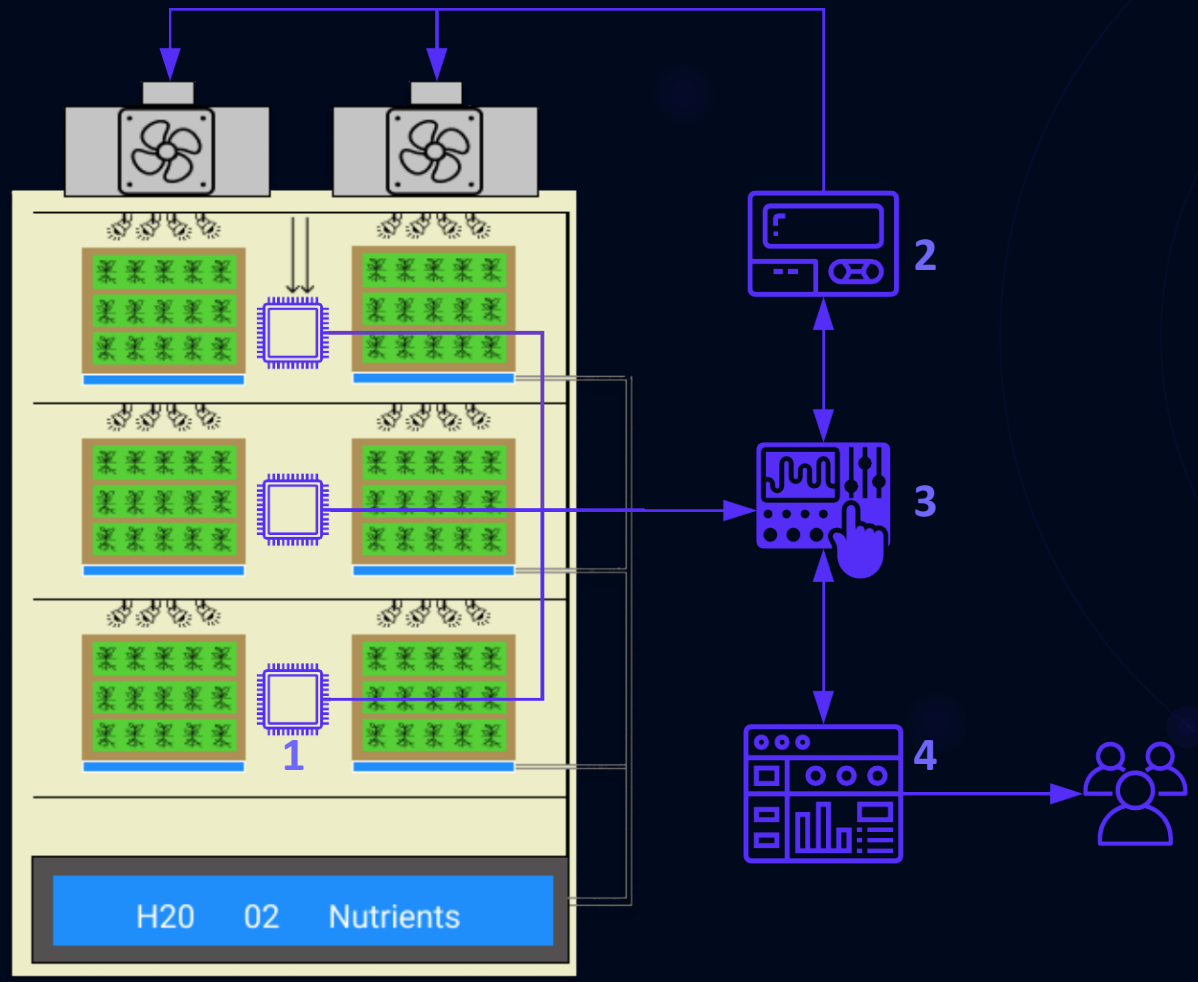
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Solution

4-Part Tech Stack

Ambient Sensors, Edge Nodes, and Controller are connected to the exiting equipment to provide an additional layer of intelligence for building optimization.

1. Sensors

Measures four core parameters: temperature, relative humidity, light intensity, and CO_2

2. [redacted]

Adapts and controls variety of equipment

3. Control Unit

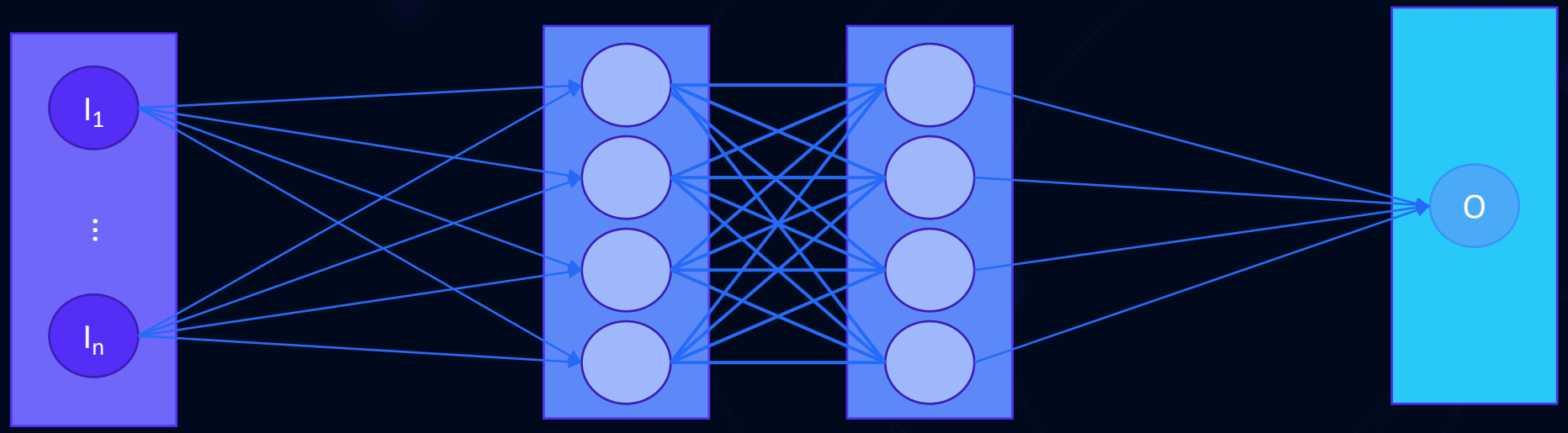
[redacted] communication among the sensor and edge nodes

4. Platform

Delivers insights and control capabilities to the user

Automation Engine

Neural Network based on Building Science



The sensors and external data are used as inputs. They are passed down with each subsequent stop or layer extracting a different set of high-level features.

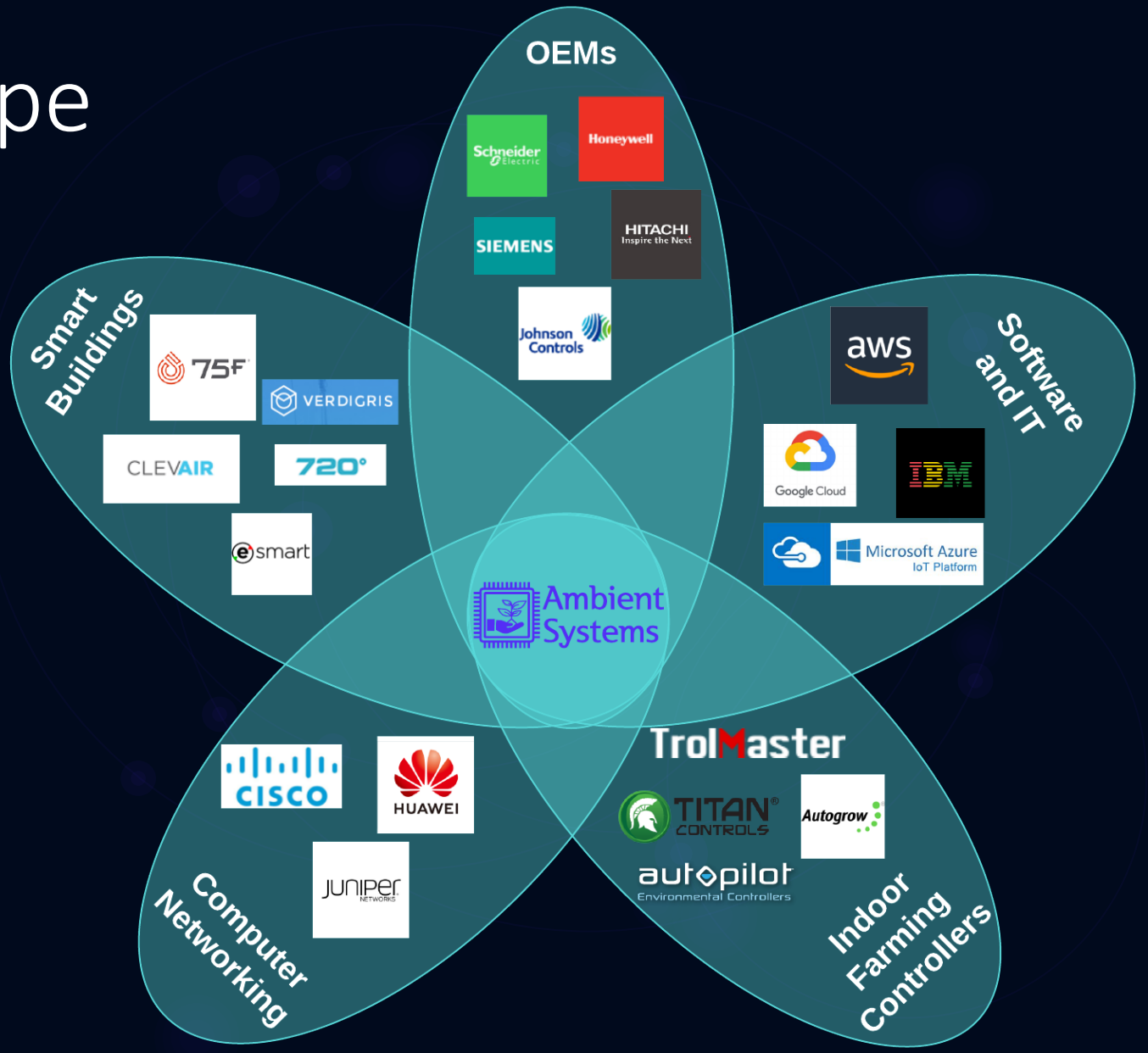
By the time fourth or fifth hidden layer is reached, deep learning net will have created complex feature detectors. It can figure out what the optimal equipment setpoints should be to achieve desired outputs.

Output Layer

Once the final output layer is reached, the farm operator will receive recommendations in real-time on what the optimal equipment set points should be for the building equipment. This can be sent directly to the site for automation.

Competitive Landscape

Ambient is positioned at the intersection of multiple segments in order to provide a tailored solution for the indoor farming beachhead market.



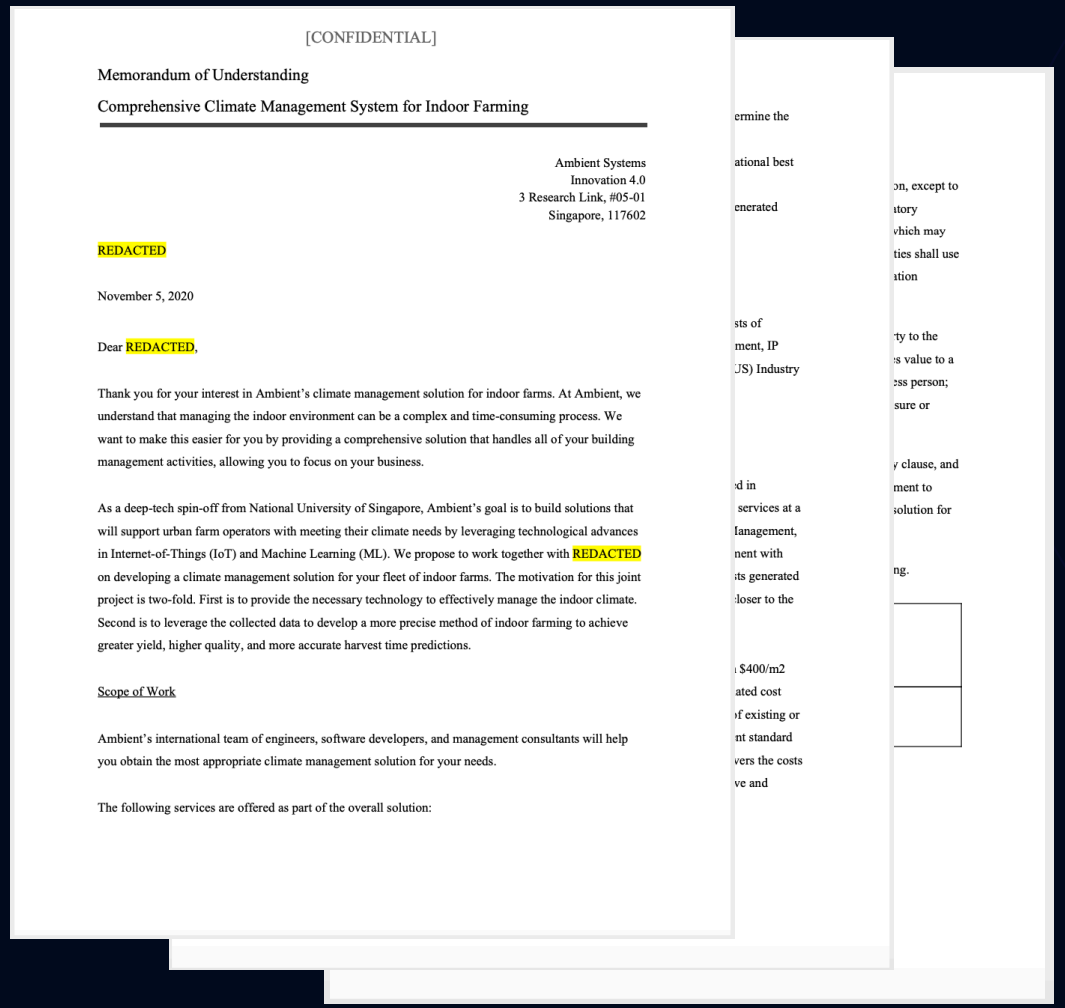
Competitive Benchmark

	Ambient	Autogrow	75F	Johnson Controls
Segment	Multiple	Indoor Farming Controller	Smart Buildings	OEM
Proximity to Customer (Low Logistics Cost)	✓ Yes	✗ No	✗ No	✓ Yes
Indoor Farming Features	✓ Yes	✓ Yes	✗ No	✗ No
Modern Web-based UI	✓ Yes	✗ No	✓ Yes	✗ Some
Wireless IoT Network	✓ Yes	✗ No	✓ Yes	✓ Yes
Advanced Optimization Methods	✓ Machine Learning	✗ No	✗ Some	✗ Some
Pricing	Quotation (based on area)	$3000 for one unit + Installation Fees	Quotation	Quotation

Innovative Revenue Model

Section	Component	Cost
CAPEX		
Services	Audit and Design	Complimentary
	Installation	$4,000 - $5,000
Hardware	Ambient Smart Sensors	$1,500 - $2,500
	Ambient Edge Nodes	$2,500 - $4,500
	Ambient Control Unit	$4,000 - $6,000
CAPEX SubTotal		**$12,000 - $18,000**
OPEX		
Operations	Secure Data Management	Complimentary
	System Maintenance	Complimentary
	Parts Replacement	Complimentary
Ambient Platform	Facility Analytics	Value-sharing
	ACMV Optimization	Value-sharing
	Predictive Maintenance	Value-sharing
	Yield Prediction	Value-sharing
	Marketplace	Value-sharing
OPEX SubTotal		**$0**
Estimated Total Cost		**$12,000 - $18,000 + OPEX value-sharing**

Customer Traction



[CONFIDENTIAL]

Memorandum of Understanding

Comprehensive Climate Management System for Indoor Farming

Ambient Systems
Innovation 4.0
3 Research Link, #05-01
Singapore, 117602

REDACTED

November 5, 2020

Dear REDACTED,

Thank you for your interest in Ambient's climate management solution for indoor farms. At Ambient, we understand that managing the indoor environment can be a complex and time-consuming process. We want to make this easier for you by providing a comprehensive solution that handles all of your building management activities, allowing you to focus on your business.

As a deep-tech spin-off from National University of Singapore, Ambient's goal is to build solutions that will support urban farm operators with meeting their climate needs by leveraging technological advances in Internet-of-Things (IoT) and Machine Learning (ML). We propose to work together with REDACTED on developing a climate management solution for your fleet of indoor farms. The motivation for this joint project is two-fold. First is to provide the necessary technology to effectively manage the indoor climate. Second is to leverage the collected data to develop a more precise method of indoor farming to achieve greater yield, higher quality, and more accurate harvest time predictions.

Scope of Work

Ambient's international team of engineers, software developers, and management consultants will help you obtain the most appropriate climate management solution for your needs.

The following services are offered as part of the overall solution:

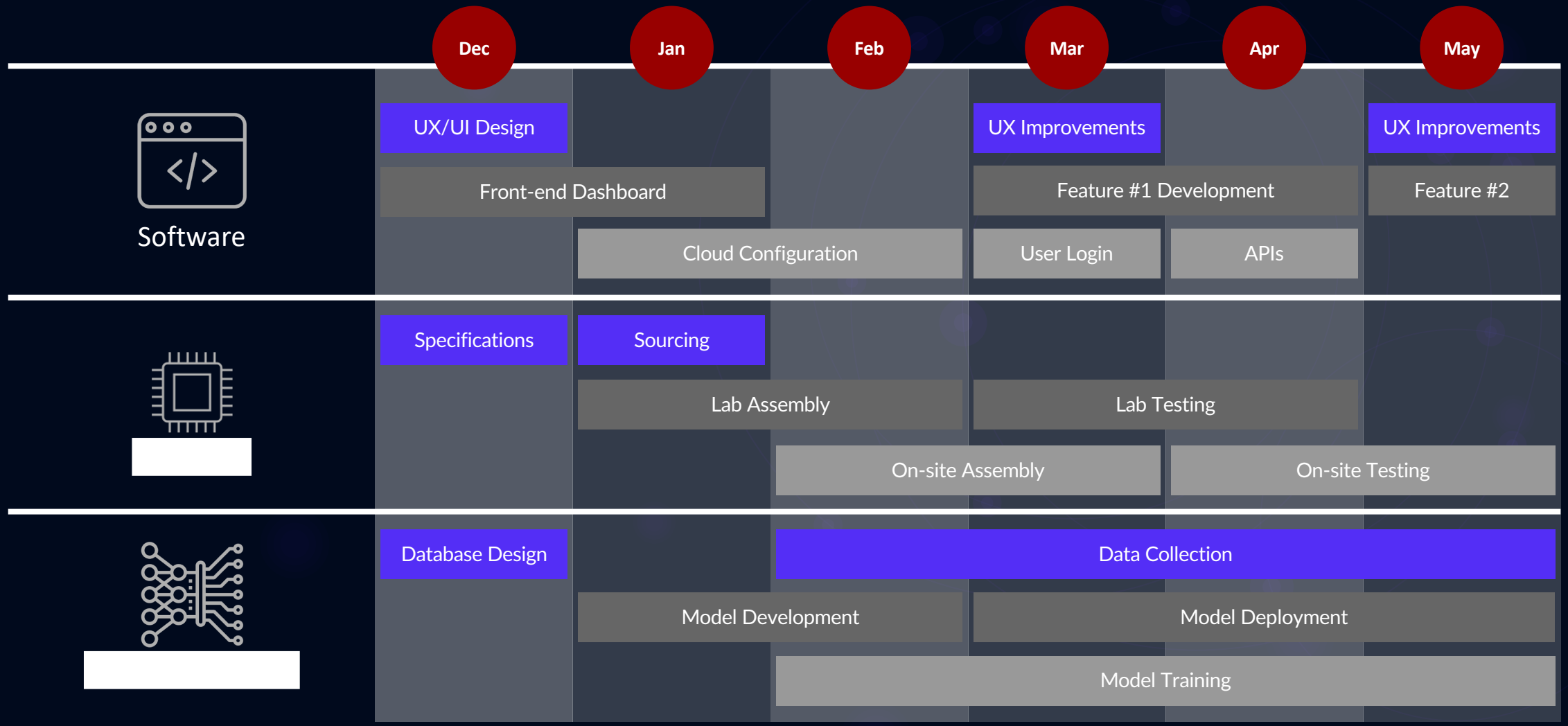
Summary:

- Signed an MoU with farm operator in Singapore
- Value sharing details to be discussed and confirmed closer to the agreement
- In conversation with two leading farm operators
- Identified gap for similar solution in the healthcare sector

The Ask

1. $10k bridge capital for the pilot
2. Advisory support from VM, TM, and Master Engineer during GRIP Run 5
3. Opportunity to pitch in GRIP Run 5 for $100k

PRODUCT ROADMAP & MILESTONES





Summary

1. Ambient has identified a competitive product-market fit as a building technology provider for indoor farms.
2. Initial customer is interested to move foreword with the pilot project.
3. Asking for an extension to pitch at GRIP Run 5 and $10k bridge capital for the pilot.

Contact: ivan@ambientsystems.ai